

## GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

|                                |                                                                |
|--------------------------------|----------------------------------------------------------------|
| <b>Project</b>                 | FEED for Gas Gathering Pipelines & Associated Infrastructure   |
| <b>Client</b>                  | Brass Fertilizer and Petrochemical Company Limited             |
| <b>Originating Company</b>     | Epic Atlantic Limited                                          |
| <b>Document Title</b>          | Specification for Pressure, Temperature and Level Transmitters |
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| <b>Originator</b>              | Paul Roy                                                       |
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| <b>Issue Date</b>              | 19-Oct-22                                                      |

## Specification for Pressure, Temperature and Level Transmitters

|       |            |                      |            |           |                 |                  |
|-------|------------|----------------------|------------|-----------|-----------------|------------------|
|       |            |                      |            |           |                 |                  |
| R02   | 19.10.22   | Re-issued for Review | P. Roy     | P. Olukah | A. Duru         |                  |
| R01   | 18.07.22   | Issued for Review    | P. Roy     | P. Olukah | A. Duru         |                  |
| Rev # | Issue Date | Status Description   | Originator | Reviewer  | Approver (Epic) | Approver (BFPCL) |

## Revision History

| Rev No. | Date     | Description of Revision                                            |
|---------|----------|--------------------------------------------------------------------|
| R01     | 18.07.22 | Issued for Review                                                  |
| R02     | 19.10.22 | Re-issued for review after addressing TCE comments on revision R01 |
|         |          |                                                                    |
|         |          |                                                                    |

## Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)  
*for the subsequent project phases, post FEED, it is expected that:*
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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## SUMMARY PAGE

|                  |                                                     |                                                       |
|------------------|-----------------------------------------------------|-------------------------------------------------------|
| <b>Client:</b>   | Brass Fertilizer & Petrochemical Company Limited    |                                                       |
| <b>Project:</b>  | Gas Processing & Methanol Complex                   | <b>Project No.</b> 11998A                             |
| <b>Facility:</b> | Gas Gathering Pipelines & Associated Infrastructure | <b>Doc No.</b><br>EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-002 |

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

**KEYWORDS:** BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

## REFERENCES

| Document No.                          | Description                                                 |
|---------------------------------------|-------------------------------------------------------------|
| EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001   | Upstream Gas Gathering Concept Select & Revalidation Report |
| EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001 | Preliminary Simulation – Based on Desktop Route Survey      |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-002      | Gas Gathering & Evacuation Pipelines Concept Screening      |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-001      | Conceptual Study Memorandum                                 |
| EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002   | Basis of Design                                             |
|                                       | FISAS Draft ESIA Report (November 2020)                     |
|                                       |                                                             |

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## Abbreviations

| Abbreviation | Meaning                         |
|--------------|---------------------------------|
| PG           | Pressure Gauge                  |
| TG           | Temperature Gauge               |
| LG           | Level Gauge                     |
| RO           | Restriction Orifice Plate       |
| RTD          | Resistance Temperature Detector |
| SPD          | Surge Protection Devices        |

# 1. Introduction

## 1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

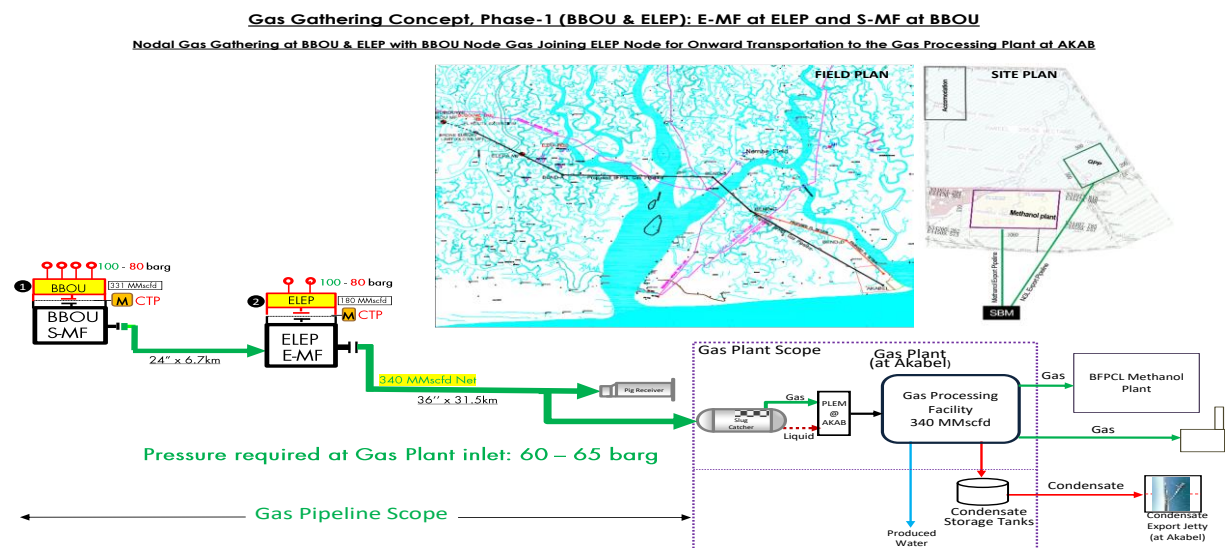


Figure 1: Gas Gathering Concept, Phase 1

## 1.2. Purpose

This document defines the basic requirements for the field Transmitter like Pressure, Temperature and Level Transmitters. The FEED shall fully comply with all specified relevant contractual requirements. This document defines the basic requirements for the engineering and design of the instrumentation for BBOU, ELEP and AKAB pipeline system. The purpose of this specification is to ensure a standard approach is maintained to the engineering and design of instrumentation and equipment and that it complies with all relevant national and international standards.

The FEED shall fully comply with all specified relevant contractual requirements.

## 2. Regulations, Codes And Standards

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The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

### 3. General

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- 3.1. Instruments and equipment shall be consistent with the environmental impact statement, with particular emphasis to the following:

Noise within 20 meters of the battery limit must not exceed 45dBA. Fugitive process emissions must be minimized e.g. from control valve glands and analyser vents.

- 3.2. Field instrumentation shall have a minimum ingress protection of IP65 in accordance with IEC 60529. Ingress protection for Level Gauge Indicator shall be IP-67. All instruments shall be intrinsically safe or explosion proof and shall be certified to specific hazardous area class by statutory body such as ATEX, PTB.

3.3. **Materials of Construction**

Materials of construction for instrument equipment shall be consistent with corrosion resistance and the pressure and temperature conditions of the process fluids involved. Generally, the parts in contact with the process fluid shall be of a material equal to or better than that used for the related vessel or line. Where instrument design prohibits the use of these materials, or where a suitable material cannot be reasonably procured, the instruments shall be isolated from the process by means of a chemical seal which shall be identified on P&ID.

Wetted parts including sensing elements and also movement parts shall be of AISI 316 stainless steel. The material employed in special services shall be selected according to application from SS316L, Titanium, Monel, Hastelloy and etc. The materials of wetted part in sour services shall generally be selected and treated in accordance with the recommendation of NACE MR 0103

The following materials shall not be used in contact with the product:

Aluminium for wetted parts, Zinc, Asbestos, Brass, Bronze, Copper. Heavy metals including Mercury, Antimony and Cadmium. Plastics containing free Phenol, formaldehyde or plasticizers.

3.4. **Pressure & temperature rating**

Pressure and temperature rating of instruments and components shall conform to the design rating of the process system to which they are connected. Instruments which can be exposed to vacuum shall have under-range protection to full vacuum

- 3.5. All transmitters shall be supplied with external Surge Protection Devices (SPDs)

3.6. **Hazardous area requirement**

- 3.6.1. Classification of Hazardous Areas shall be Zone1, IIA/IIB, T4.

All instrumentation located or partially located in the hazardous area shall be suitable for classified hazardous area. All instruments and Junction Boxes shall be certified as Explosion Proof (Exd).

Power supply shall be switched off before opening the Instrument or Junction Box. ATEX/PTB approved certification shall be supplied for all electrical / electronic equipment in the hazardous area.

## 4. Temperature Instruments

### 4.1. Transmitters

- 4.1.1. Microprocessor based, SMART (4-20 mA with HART) with local LCD indication shall be used.
- 4.1.2. Transmitter shall be integrally mounted with RTD. Accuracy of the Temperature Transmitter shall be  $\pm 0.08$  degC.
- 4.1.3. In the event of temperature element failure, transmitters shall be arranged to drive towards the alarm condition or safe control condition where one exists, or otherwise upscale.

### 4.2. Protecting Thermowells

- 4.2.1. Thermowell shall be machined from bar stock material. Other materials shall be furnished as required in the piping specification. Temperature elements shall be supplied complete with thermowell, extension, head and integrally mounted transmitter as a complete assembly. All thermowells shall be SS316L and meet the Karman Wake Frequency calculation as per PTC 19.3. If it fails to achieve the wake frequency calculation, collar or vortex shedding design shall be used to meet the velocity criteria. Thermowell design and geometry shall meet the requirements of Part II, Section 3.9 of DEP 31.38.01.24-Gen. At all process conditions, the ratio of the Strouhal frequency and the thermowell natural frequency shall be less than 0.8, as stated in ASME PTC 19.3.
- 4.2.2. Thermowells shall be flanged according to equipment and pipe specifications. The flange size shall be 40 mm (1.1/2 inch) with rating and facing according to the line classification.
- 4.2.3. According to API RP 551, the immersion length of thermowells in the pipes (the length from well tip to pipe wall) shall be maximum 5" or 125mm. The immersion length in vessels liquid shall be 400 mm. In vessels containing gas, the immersion length shall be according to the table below:

| Vessel diameter (D)    | Thermowell immersion length(mm) |
|------------------------|---------------------------------|
| D up to 800 mm         | 200                             |
| $800 < D \leq 1200$ mm | 300                             |

The insertion length depends on the length of the nozzle or pipe fitting, taking any possible refractory lining into account.

If the thermowell design fails to pass the wake frequency with the give gas velocity even after increasing the tip diameter or other mechanical parameters, vortex shedding type helical thermowell shall be used. Velocity collar shall be avoided.

## 5. Pressure Instruments

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### 5.1. Pressure transmitters

- 5.1.1. Pressure transmitters shall be referenced to atmospheric pressure.
- 5.1.2. Transmitters shall be microprocessor based smart (4-20mA HART) type, with display unit and independent adjustment of zero and span at the instrument and from a remote location. Accuracy shall be  $\pm 0.075\%$  of the calibrated span.
- 5.1.3. Materials of construction shall be suitable for the process media and based upon the piping specification. In general pressure transmitters should have Aluminium Enclosure with low copper and stainless-steel wetted parts.
- 5.1.4. Scales shall be selected to cover the range requirements for both start-up, normal and design conditions of plant operation.

### 5.2. Instrument Manifolds

- 5.2.1. 2-valve manifolds shall be used for secondary isolation, venting and calibration of pressure instruments.
- 5.2.2. 5-valve manifolds shall be used for secondary isolation, venting and calibration of differential pressure instruments
- 5.2.3. Alternative to 2-valve and 5-valve manifold, mono-flange with double block and bleed (DBB) can be used. Mono-flange shall have  $\frac{1}{2}$ " NPTF connection for pressure gauge and pressure transmitter connection. Mono-flange assembly shall be provided with closed-coupled hook-up. For unmanned platforms, mono-flange assembly shall help in reducing/elimination impulse tubing that could be source of leakage.

## 6. Level Instruments

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### 6.1. Level Transmitters

- 6.1.1. The level is measured by short radar pulses, which are transmitted from the antenna at the tank top towards the liquid. When a radar pulses reaches a media with a different dielectric constant, part of the energy is reflected back to the transmitter. The time difference between the transmitted and the reflected pulse is proportional to the distance, from which the level is calculated.
- 6.1.2. Guided Wave Radar shall be used for the measurement of level for CBD Tanks.
- 6.1.3. Guided Wave Radar shall be based on the Time Domain Reflectometry (TDR) principle. Radar Transmitter shall be of low frequency type within the range 6 GHz
- 6.1.4. It shall have rigid rod of SS316 material. PTFE sealing and Viton "O" ring shall be considered.
- 6.1.5. Accuracy shall be +/- 5 mm
- 6.1.6. The power supply for non-contacted type of radar transmitter, if used shall be 24VDC. The cable entries of radar shall be 1/2 NPT(F), 2 ports.
- 6.1.7. The process connection of guided wave radar shall be 3", 150#RF flanged. Guided wave radar shall be loop-powered.
- 6.1.8. Radar Transmitter measurement range shall be 6 meters. Dielectric Constant of the liquid shall be taken into consideration while calculating the actual tank level. Beam angle and antenna diameter shall be accordingly selected.
- 6.1.9. Output signal shall be 4-20mA with HART communication and shall have integral LCD display. Radar transmitter shall two wire type and power supply (24VDC) shall be supplied to transmitter using same wire of communication.

## 7. Tag List

| Sr. No. | Tag No.       | Type of Instrument            | Line Rating | Line Pressure Range | Accessories                                  |
|---------|---------------|-------------------------------|-------------|---------------------|----------------------------------------------|
| 1       | BBOU-PIT-1001 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 2       | BBOU-TIT-1001 | Temperature Transmitter       | #900        | 0-100 degC          | M20 Gland, External SPD                      |
| 3       | BBOU-PIT-1002 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 4       | BBOU-PIT-1003 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 5       | BBOU-PIT-1004 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 6       | BBOU-PIT-1005 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 7       | BBOU-PIT-1013 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 8       | BBOU-PIT-1014 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 9       | BBOU-PIT-1015 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 10      | BBOU-PIT-1016 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 11      | BBOU-LIT-1001 | Guided Wave Radar Transmitter | #150        | 0-2400mm            | M20 Gland, External SPD                      |
| 12      | BBOU-TIT-1003 | Temperature Transmitter       | #900        | 0-100 degC          | M20 Gland, External SPD                      |
| 13      | BBOU-LIT-1002 | Guided Wave Radar Transmitter | #150        | 0-2400mm            | M20 Gland, External SPD                      |
| 14      | ELEP-DPT-1001 | DP Transmitter                | #900        | 0- 5000 mmWC        | Monoflange Manifold, M20 Gland, External SPD |
| 15      | ELEP-DPT-1002 | DP Transmitter                | #900        | 0- 5000 mmWC        | Monoflange Manifold, M20 Gland, External SPD |
| 16      | ELEP-PIT-1001 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 17      | ELEP-TIT-1001 | Temperature Transmitter       | #900        | 0-100 degC          | M20 Gland, External SPD                      |
| 18      | ELEP-PIT-1002 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 19      | ELEP-PIT-1003 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 20      | ELEP-PIT-1004 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 21      | ELEP-PIT-1005 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 22      | ELEP-PIT-1010 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 23      | ELEP-TIT-1003 | Temperature Transmitter       | #900        | 0-100 degC          | M20 Gland, External SPD                      |
| 24      | ELEP-PIT-1018 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 25      | ELEP-PIT-1019 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 26      | ELEP-PIT-1020 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 27      | ELEP-PIT-1021 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |

| Sr. No. | Tag No.       | Type of Instrument            | Line Rating | Line Pressure Range | Accessories                                  |
|---------|---------------|-------------------------------|-------------|---------------------|----------------------------------------------|
| 28      | ELEP-LIT-1001 | Guided Wave Radar Transmitter | #150        | 0-2400mm            | M20 Gland, External SPD                      |
| 29      | ELEP-TIT-1004 | Temperature Transmitter       | #900        | 0-100 degC          | M20 Gland, External SPD                      |
| 30      | ELEP-LIT-1002 | Guided Wave Radar Transmitter | #150        | 0-2400mm            | M20 Gland, External SPD                      |
| 31      | AKAB-PIT-1001 | Pressure Transmitter          | #900        | 0-100 barg          | Monoflange Manifold, M20 Gland, External SPD |
| 32      | AKAB-TIT-1001 | Temperature Transmitter       | #900        | 0-100 degC          | M20 Gland, External SPD                      |

Tagwise datasheets including process parameters shall be provided during Detailed Engineering Design when vendor inputs have been obtained.